Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, February/March - 2022
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) How do you optimize the Backup process for a Data Warehouse?
b) List and Explain briefly the techniques used to implement a multidimensional view in data warehouse. [7+8]

2.a) Describe the Data Warehouse architecture.
b) Distinguish between OLTP and OLAP systems. [7+8]

3.a) Illustrate the various data reduction techniques for data preprocessing.
b) List the steps of the Knowledge Discovery in Databases (KDD) and describe each of them. [7+8]

4.a) Discuss in brief about Data Mining Issues.
b) Discuss about discretization and concept hierarchy generation for numerical data. [7+8]

5.a) How are association rules generated from frequent itemsets? Illustrate.
b) Explain the procedure to mining closed frequent data item sets. [8+7]

6.a) Apply FP-growth algorithm on the following database to find all of the strong association rules with min_sup = 60% and min_conf = 80%.

TID	items
T100	I1, I2, I3, I4, I5, I6
T200	I2, I3, I4, I5, I6, I7
T300	I1, I4, I5, I8
T400	I1, I4, I6, I9, I11
T500	I2, I4, I5, I9, I10

b) Can we design a method that mines the complete set of frequent item sets without candidate generation? Explain. [8+7]

7.a) Discuss on Measures for Selecting the Best Split for the various types of attributes.
b) Explain k-NN Algorithm for data classification with an example. [8+7]

8.a) How to cluster the data sets using k-mediod clustering algorithm?
b) Discuss about classification of outlier detection techniques. [8+7]

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